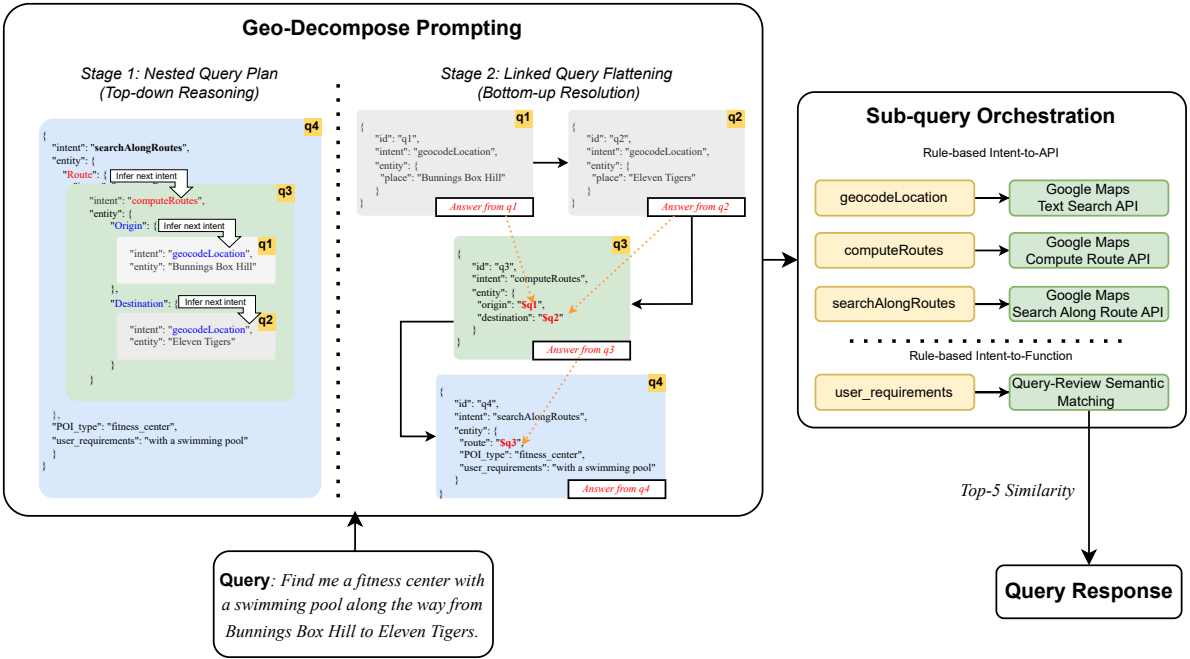
Geo-Decompose Prompting
Stage 1: Nested Query Plan
(Top-down Reasoning)
Stage 2: Linked Query Flattening
(Bottom-up Resolution)
q4
"intent": "searchAlongRoutes",
"entity": {
"Route": {
Infer next intent
q3
"intent": "computeRoutes",
"entity": {
"Origin": {
Infer next intent
q1
"intent": "geocodeLocation",
"entity": "Bunnings Box Hill"
},
"Destination": {
Infer next intent
q2
"intent": "geocodeLocation",
"entity": "Eleven Tigers"
}
}
},
"POI_type": "fitness_center",
"user_requirements": "with a swimming pool"
}
}
q1
"id": "q1",
"intent": "geocodeLocation",
"entity": {
"place": "Bunnings Box Hill"
}
}
Answer from q1
q2
"id": "q2",
"intent": "geocodeLocation",
"entity": {
"place": "Eleven Tigers"
}
}
Answer from q2
q3
"id": "q3",
"intent": "computeRoutes",
"entity": {
"origin": "Sq1",
"destination": "Sq2"
}
}
Answer from q3
q4
"id": "q4",
"intent": "searchAlongRoutes",
"entity": {
"route": "Sq3",
"POI_type": "fitness_center",
"user_requirements": "with a swimming pool"
}
}
Answer from q4
Sub-query Orchestration
Rule-based Intent-to-API
geocodeLocation
Google Maps Text Search API
computeRoutes
Google Maps Compute Route API
searchAlongRoutes
Google Maps Search Along Route API
Rule-based Intent-to-Function
user_requirements
Query-Review Semantic Matching
Top-5 Similarity
Query Response
Query: Find me a fitness center with a swimming pool along the way from Bunnings Box Hill to Eleven Tigers.